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08-04-2026

Case Study:

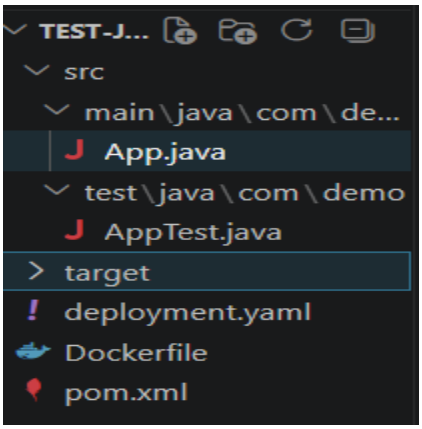
Smart Electricity Billing System:

Problem Statement:

- Ensure electricity usage is calculated and billed correctly

Objective

To develop a system that calculates electricity bills accurately based on usage and ensures correctness through automated testing, while integrating CI/CD for automated build, test, and deployment.



1.Develop Billing calculation logic:

App.java:

```
package com.demo;
public class App {
    public static void main(String[] args) {
        int units = 120;
        double bill = calculateBill(units);
        System.out.println("Units: " + units);
        System.out.println("Total Bill: " + bill);
        while(true){
            try {
                Thread.sleep(10000);
            } catch(Exception e){}
        }
    }
    public static double calculateBill(int units) {
        double bill = 0;
        if (units <= 100) {
            bill = units * 1.5;
        } else if (units <= 200) {
            bill = (100 * 1.5) + ((units - 100) * 2.5);
        } else {
            bill = (100 * 1.5) + (100 * 2.5) + ((units - 200) * 4);
```

```
        }
        return bill;
    }
}
```

2.Write Junit tests for tariff calculation:

AppTest.java:

```
package com.demo;
import static org.junit.Assert.assertEquals;
import org.junit.Test;
public class AppTest {
    @Test
    public void testBillCalculation() {
        double result = App.calculateBill(150);
        assertEquals(275.0, result, 0.01);
    }
}
```

Deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: java-app
  template:
    metadata:
      labels:
        app: java-app
    spec:
      containers:
      - name: java-container
        image: sanjayioniq/java-app:v3
        imagePullPolicy: Always
        ports:
        - containerPort: 8080
---
apiVersion: v1
```

```
kind: Service
metadata:
  name: java-service
spec:
  type: NodePort
  selector:
    app: java-app
  ports:
    - port: 80
      targetPort: 8080
      nodePort: 30007
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.demo</groupId>
    <artifactId>java-app</artifactId>
    <version>1.0</version>
    <dependencies>
        <dependency>
```

```
        <groupId>junit</groupId>
        <artifactId>junit</artifactId>
        <version>4.13.2</version>
        <scope>test</scope>
    </dependency>
</dependencies>
<build>
    <plugins>
        <!-- Compile Java -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.8.1</version>
            <configuration>
                <source>8</source>
                <target>8</target>
            </configuration>
        </plugin>
        <!-- Make executable jar -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-jar-plugin</artifactId>
            <version>3.2.0</version>
```

```
            <configuration>
                <archive>
                    <manifest>
                        <mainClass>com.demo.App</mainClass>
                    </manifest>
                </archive>
            </configuration>
        </plugin>
    </plugins>
  </build>
</project>
```

Docker.file

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "app.jar"]
```

4.Deploy microservice on Docker & kubernetes:

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> mvn clean test
```

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.084 s -- in com.demo.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.362 s
[INFO] Finished at: 2026-04-15T17:54:31+05:30
[INFO] ------------------------------------------------------------------------
```

mvn clean package(Build Maven):

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------------< com.demo:java-app >-------------------------
[INFO] Building java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
```

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.083 s -- in com.demo.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.0:jar (default-jar) @ java-app ---
[INFO] Building jar: C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app\target\java-app-1.0.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.980 s
[INFO] Finished at: 2026-04-15T17:55:22+05:30
```

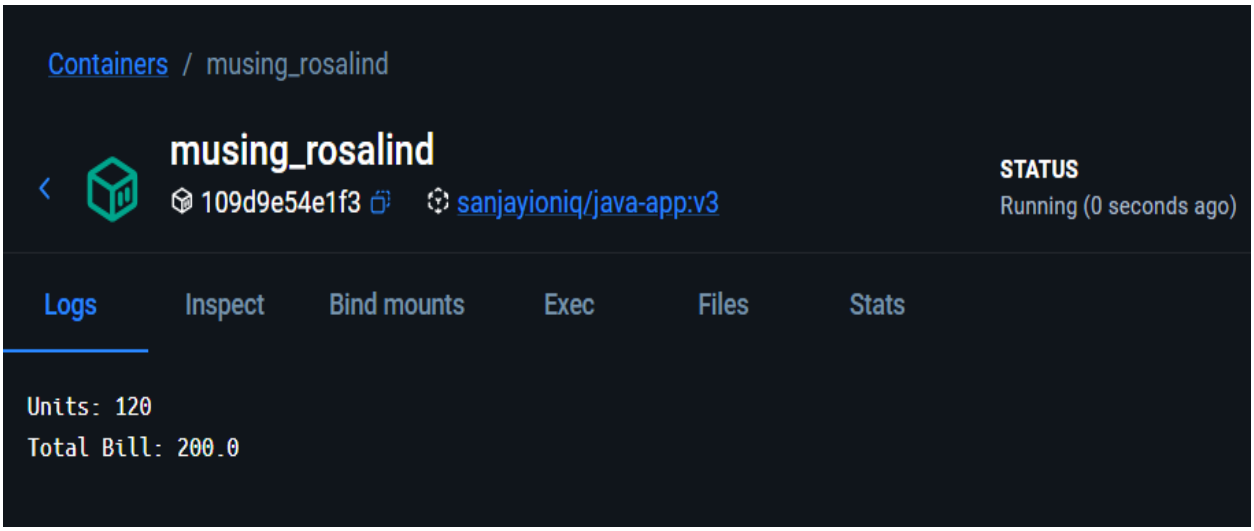
Docker

docker build -t sanjayioniq/java-app .v3 :

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> docker build -t sanjayioniq/java-app:v3 .
[+] Building 2.9s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 167B
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec4745125a47bd4f882
 => => resolve docker.io/library/eclipse-temurin:17-jdk-alpine@sha256:903e865ecd125a01759fd156d042b998aa86957262f6dec4745125a47bd4f882
 => [internal] load build context
 => => transferring context: 2.87kB
 => CACHED [2/3] WORKDIR /app
 => [3/3] COPY target/*.jar app.jar
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:b5ff0e20f7d22d4365307110a50d6aee8fad1b68c63184f8e4a1a3216eb998cf
 => => exporting config sha256:194ceb1feb17891e3e0786808caa56ae2c5ae34009702ea2a136867ca4046044
 => => exporting attestation manifest sha256:fe78415514ed4efe452c3e80ba431f08b014800ea6d2113085dea97f2473ae56
 => => exporting manifest list sha256:82f8ab90ad37e3cf64645c7dc1e0a4a3ba21ffbc1dac3fa9eb51304fdce3f401
 => => naming to docker.io/sanjayioniq/java-app:v3
 => => unpacking to docker.io/sanjayioniq/java-app:v3
```

docker run sanjayioniq/java-app:v3

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> docker run sanjayioniq/java-app:v3
Units: 120
Total Bill: 200.0
```



docker push sanjayioniq/java-app:v3

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> docker push sanjayioniq/java-app:v3
The push refers to repository [docker.io/sanjayioniq/java-app]
75d716ac6ad3: Layer already exists
82c540078c28: Layer already exists
10ac7a5474fd: Pushed
1e15b1894c9c: Pushed
589002ba0eae: Layer already exists
7cd2436f004e: Layer already exists
3a547da4fe15: Layer already exists
946c51efe0fd: Layer already exists
v3: digest: sha256:a81d086d0551efbea691161486f7baddba6a4bd7d27c7046aebbe6608a0fbf05 size: 856
```

KUBERNETES:

kubectl apply -f deployment.yaml:

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> kubectl apply -f deployment.yaml
>>
deployment.apps/java-app unchanged
service/java-service unchanged
```

Kubectl get pods & service:

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
java-app-9644b5f77-gvdtr    1/1     Running   0          2m45s
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> kubectl get service
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
java-service   NodePort    10.96.149.85   <none>        80:30007/TCP   7d21h
kubernetes     ClusterIP   10.96.0.1      <none>        443/TCP        15d
```

KUBERNETES:

Kubernetes Give feedback

Namespace
default

Cluster

Stop | Edit cluster

Cluster	Cluster type	Nodes	Kubernetes version
Active	kind	1	v1.34.3

Deployments

Name	Status	Pods
java-app	Available	1/1

Pods

Name	Status
java-app-9644b5f77-gvdtr	Running

Nodes

Name	Status
desktop-control-plane	Ready

Services

Name	Cluster IP	Ports
java-service	10.96.149.85	80/TCP
kubernetes	10.96.0.1	443/TCP

CONTAINERS:

Containers Give feedback

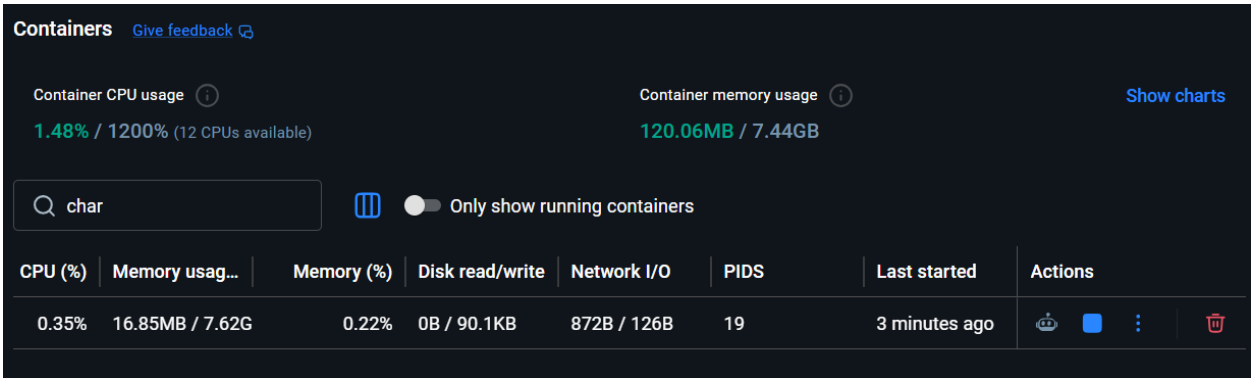
Container CPU usage
1.28% / 1200% (12 CPUs available)

Container memory usage
120.06MB / 7.44GB

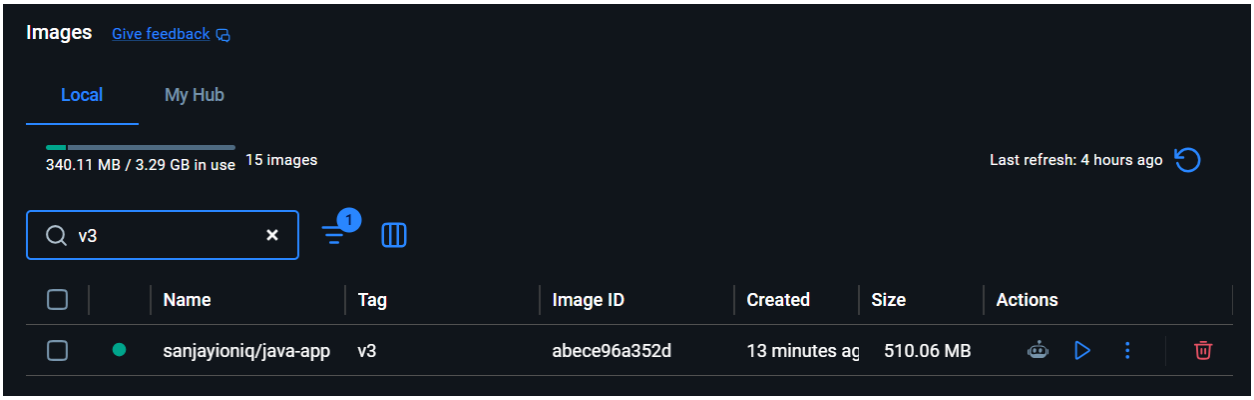
char

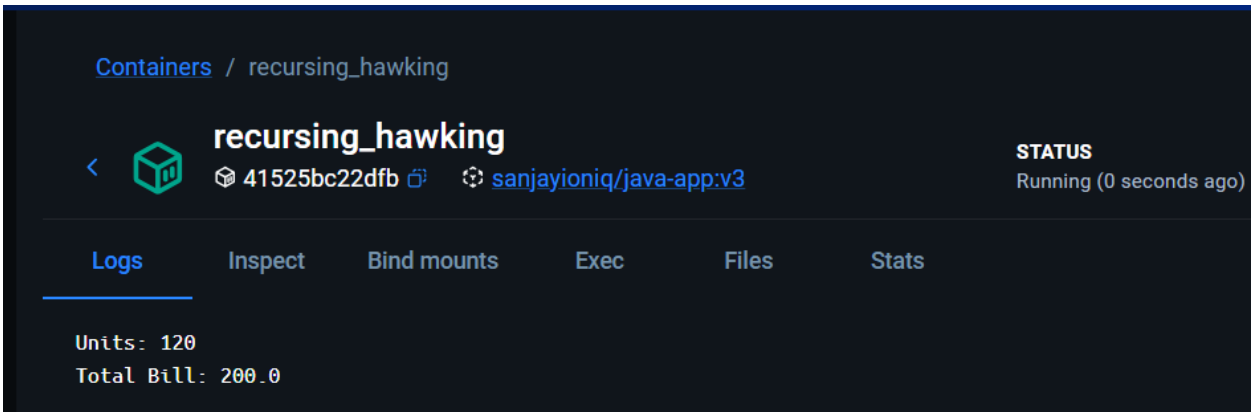
Only show running containers

Name	Container ID	Image	Port(s)
charming_wozn	b5f722538552	sanjayionic	


Containers
Give feedback
Container CPU usage
1.48% / 1200% (12 CPUs available)
Container memory usage
120.06MB / 7.44GB
Show charts
char
Only show running containers
CPU (%)
Memory usag...
Memory (%)
Disk read/write
Network I/O
PIDS
Last started
Actions
0.35%
16.85MB / 7.62G
0.22%
0B / 90.1KB
872B / 126B
19
3 minutes ago

IMAGES:


Images
Give feedback
Local
My Hub
340.11 MB / 3.29 GB in use
15 images
Last refresh: 4 hours ago
v3
Name
Tag
Image ID
Created
Size
Actions
sanjayioniq/java-app
v3
abece96a352d
13 minutes ag
510.06 MB

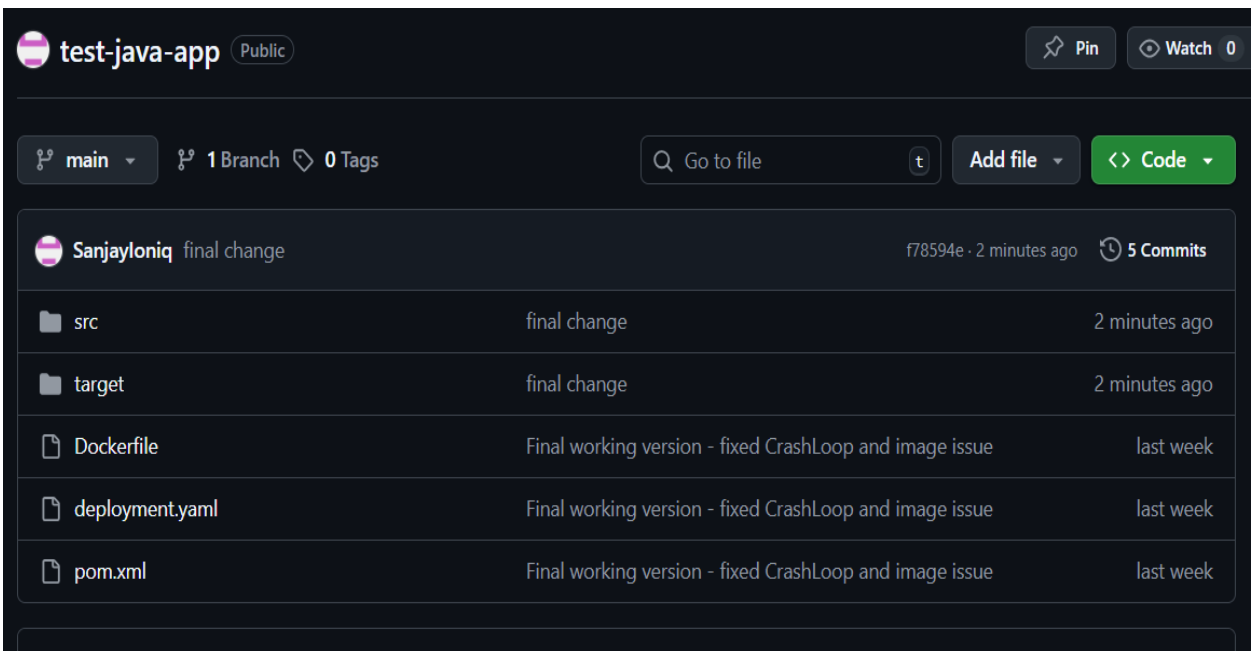

Containers / recursing_hawking
recursing_hawking
41525bc22dfb
sanjayioniq/java-app:v3
STATUS
Running (0 seconds ago)
Logs
Inspect
Bind mounts
Exec
Files
Stats
Units: 120
Total Bill: 200.0

4.Implement CI/CD pipeline using automated testing

Git Push:

```
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> git add  .
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> git commit -m "final change"
[main f78594e] final change
 7 files changed, 24 insertions(+), 49 deletions(-)
PS C:\Users\sanja\Sanjay Balu\OneDrive\Documents\test-java-app> git push -u origin main
Enumerating objects: 50, done.
Counting objects: 100% (50/50), done.
Delta compression using up to 12 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (26/26), 4.12 KiB | 703.00 KiB/s, done.
Total 26 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/SanjayIoniq/test-java-app.git
   aea808d..f78594e  main -> main
branch 'main' set up to track 'origin/main'.
```

Git status:



Jenkins-Pipeline:

Script:

```
pipeline {
    agent any
    environment {
        IMAGE_NAME = "sanjayioniq/java-app"
        IMAGE_TAG = "v${BUILD_NUMBER}"   // auto version
        FULL_IMAGE = "${IMAGE_NAME}:${IMAGE_TAG}"
    }
    stages {
        stage('Clone Repository') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/sanjayioniq/test-java-app.git'
            }
        }
        stage('Build & Test') {
            steps {
                bat 'mvn clean test'
            }
            post {
                always {
                    junit 'target/surefire-reports/*.xml'
                }
```

```
        }
    }
    stage('Package') {
        steps {
            bat 'mvn package -DskipTests'
        }
    }
    stage('Build Docker Image') {
        steps {
            bat "docker build -t %FULL_IMAGE% ."
        }
    }
    stage('Push to DockerHub') {
        steps {
            withCredentials([usernamePassword(
                credentialsId: 'docker-cred',
                usernameVariable: 'sanjayioniq',
                passwordVariable: 'dckr_pat_0i7j2Z-BpSsDAFlUwCIqecwOfIo'
            )]) {
                bat """
                echo %DOCKER_PASS% | docker login -u %DOCKER_USER% --
password-stdin
                docker push %FULL_IMAGE%
                """
            }
```

```
        }
    }
    stage('Deploy to Kubernetes') {
        steps {
            withCredentials([file(credentialsId: 'kubeconfig', variable:
'KUBECONFIG_FILE')]) {
                bat """
                set KUBECONFIG=%KUBECONFIG_FILE%
                kubectl apply -f deployment.yaml
                kubectl set image deployment/java-app java-
container=%FULL_IMAGE%
                kubectl rollout restart deployment java-app
                """
            }
        }
    }
    stage('Verify Deployment') {
        steps {
            withCredentials([file(credentialsId: 'kubeconfig', variable:
'KUBECONFIG_FILE')]) {
                bat """
                set KUBECONFIG=%KUBECONFIG_FILE%
                kubectl get pods
                kubectl get services
                """
```

```
            }
        }
    }
  }
}
```

Output:

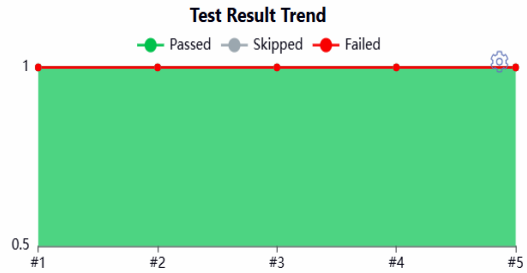
```
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ java-app ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.demo.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.066 s -- in com.demo.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.449 s
[INFO] Finished at: 2026-04-15T18:26:51+05:30
[INFO] ------------------------------------------------------------------------
Post stage
[Pipeline] junit
```

```
[INFO] --- jar:3.2.0:jar (default-jar) @ java-app ---
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\final-cicd -fat\target\java-app-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.220 s
[INFO] Finished at: 2026-04-15T18:26:56+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat


C:\ProgramData\Jenkins\.jenkins\workspace\final-cicd -fat>set KUBECONFIG=****

C:\ProgramData\Jenkins\.jenkins\workspace\final-cicd -fat>kubectl get pods
NAME                        READY   STATUS              RESTARTS   AGE
java-app-587d8ddb46-qg9tv   1/1     Running             0          86s
java-app-7dd58c7848-m2tpc   0/1     ContainerCreating   0          0s
java-app-969d66b7c-7252c    0/1     Terminating         0          1s

C:\ProgramData\Jenkins\.jenkins\workspace\final-cicd -fat>kubectl get services
NAME           TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
java-service   NodePort    10.96.149.85   <none>        80:30007/TCP   7d21h
kubernetes     ClusterIP   10.96.0.1      <none>        443/TCP        15d
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Stage View:

final-cicd -fat

Add description



Stage View

Stage View

	Clone Repository	Build & Test	Package	Build Docker Image	Push to DockerHub	Deploy to Kubernetes	Verify Deployment
Average stage times: (full run time: ~34s)	2s	8s	5s	4s	9s	1s	873ms
#4 Apr 15 18:26 No Changes	2s	6s	4s	3s	9s	1s	886ms
#3 Apr 15 18:25 No Changes	2s	6s	4s	3s	9s	1s	837ms
#2 Apr 15 18:24 No Changes	2s	6s	5s	3s	8s	1s	899ms

Result

- The system correctly calculated electricity bills based on unit consumption.
- Duplicate errors and incorrect billing were avoided.
- Automated CI/CD pipeline ensured smooth build and deployment.
- Application was successfully deployed and running in Kubernetes.

CONCLUSION

- The Smart Electricity Billing System was successfully developed and deployed using modern DevOps practices.
- The system ensures accurate billing through slab-based calculation and validation using unit tests.
- By integrating CI/CD tools like Jenkins, Docker, and Kubernetes, the application achieved automation, scalability, and reliability.
- This project demonstrates how DevOps enhances software delivery by reducing manual effort and improving efficiency.